

NovaTech Solutions

Revenue Intelligence Dashboard

Project Submission Documentation

Student: Vivek Kumar
Program: Future AWS Agentic AI Business Professional
Date: September 20, 2026
Platform: Amazon QuickSight (SPICE)

Data Verification Log

Purpose: Verify data quality and completeness across all three indexed knowledge bases using Quick Chat natural language queries, compared against the data dictionary.

Instructions

Each entry below documents: the question asked in Quick Chat, the expected answer from the data dictionary, Q's actual response, and a Pass/Fail assessment.

#	Knowledge Base	Question Asked	Expected Answer	Q's Actual Answer	Match?
1	NovaTech CRM Deals	How many rows are in the CRM deals dataset?	499 rows	499 rows	Yes
2	NovaTech CRM Deals	How many deals were Won vs Lost?	Won: 315, Lost: 184	Won: 315, Lost: 184	Yes
3	NovaTech Marketing Campaigns	How many rows are in the marketing campaigns dataset?	2,240 rows	2,240 rows	Yes
4	NovaTech Marketing Campaigns	How many distinct campaign names are there?	6 campaigns	6 campaigns	Yes
5	NovaTech Support Tickets	How many rows are in the support tickets dataset?	3,000 rows	3,000 rows	Yes
6	NovaTech Support Tickets	How are tickets distributed by priority?	low: 1500, medium: 1050, high: 400, critical: 50	low: 1500, medium: 1050, high: 400, critical: 50	Yes
7	NovaTech CRM Deals	What is the date range of deal_closed_date?	Dec 2023 to Jan 2025	Dec 3, 2023 to Jan 31, 2025	Yes

Cross-Check Verification

Fact verified: Total number of Won deals = 315
Chat said: 315 deals with deal stage = "Won"
QuickSight shows: 315 rows when filtering by deal.stage = Won
Consistent? Yes

Data Transformation Notes

Data Type Correction

Dataset: novatech_support_tickets
Field corrected: ticket resolved date
Original type: String
Corrected type: DateTime
Why: Required for resolution time calculations

The field contains timestamps in YYYY-MM-DD HH:MM:SS format. Without correcting to DateTime, resolution time (hours between ticket creation and resolution) cannot be computed.

Calculated Fields

Field 1: Days to Close

Dataset: novatech_crm_deals
Formula: `dateDiff({ deal_created_date }, { deal_closed_date }, "DD")`
Purpose: Sales cycle length analysis
Result: Won deals avg 71.9 days; Lost deals avg 57.9 days

Field 2: Campaign ROI

Dataset: novatech_marketing_campaigns
Formula: `(( { revenue_attributed } - { campaign_spend }) / { campaign_spend })`
Purpose: Campaign profitability analysis
Result: All campaigns negative; best is NovaPulse Launch at -83.7%

Join Configuration : Unified Dataset

Dataset name: novatech_unified
Anchor table: novatech_crm_deals (499 rows)
Join 1: CRM LEFT JOIN Marketing on `account_id = account_id`
Join 2: Result LEFT JOIN Support on `account_id = account_id`
Join rationale: LEFT JOIN preserves all 499 CRM deals even without marketing/support matches

All Four Datasets Saved to SPICE

#	Dataset Name	Rows	Notes
1	novatech_crm_deals.csv	499	+ Days to Close field
2	novatech_marketing_campaigns.csv	2,240	+ Campaign ROI field
3	novatech_support_tickets.csv	3,000	DateTime fix applied
4	novatech_unified	>499	LEFT JOIN of all 3

Q Exploration Log

Student Name: Kasmia Bhatia **Date:** September 12, 2026

#	Question Asked	Q's Answer	Dashboard Visual	Dashboard Shows	Match?
1	Which campaign channel has the highest conversion rate?	Direct Mail: 53%, Paid Social: 40.3%, Partner Referral: 34.3%	Marketing Funnel — Conversion Rate by Channel	Direct Mail 53%	Yes
2	What is the average deal size by company size?	Enterprise accounts have highest avg deal value	Sales Pipeline — Revenue by Company Size	Enterprise highest	Yes
3	What is the average resolution time for critical vs low priority tickets?	Critical: 56.6 hrs; Low: 59.4 hrs — nearly identical	Customer Health — Resolution Time by Priority	Same pattern visible	Yes
4	What are the top accounts by support ticket volume and their deal revenue?	ACCT-041: 334 tickets, \$40,722 revenue — highest risk	Customer Health — At Risk Accounts table	ACCT-041 top row	Yes
5	Are there campaigns where spend exceeded revenue?	Yes — all 6 campaigns. NovaPulse Launch best at —83.7% ROI	Marketing Funnel — Spend vs Revenue chart	All bars negative	Yes

Reflection

Where did Q agree with the dashboard?

Q agreed on all five questions after Topic configuration. Row counts, conversion rates, priority distributions, and resolution times all matched exactly.

Where did Q disagree or struggle?

Before Topic configuration, Q struggled with cross-dataset questions and ROI formulas. After configuring the Topic with field descriptions and synonyms, accuracy improved significantly.

When to use Q vs dashboard?

Use the **dashboard** for weekly recurring reporting. Use **Q** for ad-hoc questions that arise mid-meeting, especially cross-dataset queries not pre-built into charts.

Executive Report for Sarah Chen

To: Sarah Chen, VP of Revenue Operations

From: Business Intelligence Analyst

Date: September 12, 2026

Re: NovaTech Revenue Intelligence Dashboard

1. Data Strategy

NovaTech's revenue data resided in three isolated systems: CRM deals (499 rows), marketing campaigns (2,240 rows), and support tickets (3,000 rows). Each shared a common `account_id` field used as the primary join key.

Data preparation steps included: importing all three CSVs into SPICE, correcting one data type issue (`ticket_resolved_date` from String to DateTime), creating two calculated fields (Days to Close and Campaign ROI), and building a unified dataset joining all three sources via LEFT JOIN anchored to CRM deals.

Orphan accounts (ACCT-101 through ACCT-115) in marketing and support that have no CRM record appear as nulls in the unified view, this is expected and documented.

2. Dashboard Design Rationale

The dashboard contains three sheets, each answering a distinct business question:

- **Marketing Funnel** — “Where is our marketing investment going and is it working?” Features campaign spend KPI, funnel chart, conversion by channel, and spend vs. revenue combo chart.
- **Sales Pipeline** — “How is our sales team performing?” Features won revenue (\$707,201), win rate (63.1%), average deal value (\$2,245), loss reasons, and days-to-close trend.
- **Customer Health** — “Which accounts are at risk?” Uses the unified dataset. Features ticket volume by account, resolution time by priority, sentiment breakdown, and at-risk accounts table.

3. How Topic Configuration Affected Q Accuracy

Before Topic configuration, Q answered single-dataset questions correctly but struggled with cross-dataset queries and business formulas. After configuring the Topic with field descriptions, synonyms (“closed deal” = Won, “ROI” = revenue/spend formula), and named filters, Q returned accurate answers to all five exploration questions including cross-dataset ones.

4. Key Insights and Recommended Actions

1. **All six campaigns are unprofitable.** Overall ROI is -90.9% . NovaEdge Awareness is worst at -97.7% . *Action: Pause NovaEdge Awareness. Reallocate to Direct Mail and NovaPulse Launch.*

2. **Direct Mail converts at 53%** : highest of any channel despite lowest volume. *Action: Scale Direct Mail spend 2–3x in Q1.*
3. **ACCT-041 is highest risk.** 334 tickets + \$40,722 revenue + negative sentiment. *Action: Assign dedicated CSM immediately. Schedule executive business review this month.*
4. **Critical tickets not resolved faster than low-priority.** All priorities average ~58 hours. *Action: Audit SLA routing rules. Set alerts for critical tickets exceeding 8 hours.*
5. **47% of losses are “No Decision Made” or “Poor Product Fit” : not competitor wins.** *Action: Add mandatory qualification gate at Opportunity stage.*

5. AI vs Dashboard Comparison

The dashboard is better for weekly recurring reporting : fast, always available, no query needed. Q is better for ad-hoc mid-meeting questions across datasets. The one area Q underperforms is visual trend analysis, it returns numbers but cannot show a trend line. **Recommendation:** Use dashboard every Monday; use Q for ad-hoc questions throughout the week.

Dashboard Annotations

The following annotations are placed on the dashboard sheets. Each includes a quantified finding, business implication, and recommended action.

Sheet 1: Marketing Funnel

Annotation 1 : Campaign ROI Alert

Placed on: Campaign Spend vs. Revenue Attributed chart

⇒ **All 6 campaigns are running at negative ROI.** Total spend: \$12.36M vs. \$1.13M revenue attributed (overall ROI: -90.9%). NovaEdge Awareness is the worst performer at -97.7% ROI (\$1.52M spent, only \$34K returned). **Recommendation: Pause NovaEdge Awareness and redirect budget to Direct Mail and NovaPulse Launch.**

Annotation 2 : Top Converting Channel

Placed on: Conversion Rate by Channel chart

⇒ **Direct Mail converts at 53% — highest of any channel.** Despite only 149 total leads, 79 responded positively. Email (288 leads) converts at just 8.7%. **Recommendation: Scale Direct Mail spend in Q1 and test at 2–3x current volume.**

Sheet 2: Sales Pipeline

Annotation 3 : Win Rate and Loss Reason

Placed on: Loss Reason chart

⇒ **Win rate is 63.1%, but 47% of losses are to inertia — not competitors.** Only 34 deals (18.5%) went to a competitor. No Decision Made and Poor Product Fit together = 86 lost deals. **Recommendation: Add a mandatory product-fit checkpoint at the Opportunity stage.**

Annotation 4 : Days to Close Insight

Placed on: Average Days to Close KPI

⇒ **Won deals take 71.9 days vs 57.9 days for lost deals.** Longer cycles correlate with wins. **Recommendation: Do not rush deals. Focus on sustained 60–90 day engagement.**

Sheet 3: Customer Health

Annotation 5 : At-Risk Account Alert

Placed on: Top Accounts by Ticket Volume table

⇒ **ACCT-041 is the highest-risk account.** \$40,722 revenue (highest in portfolio) + 334 support tickets (nearly double the next account) + negative sentiment. **Recommendation: Assign dedicated CSM to ACCT-041 this week. Schedule executive review before any upsell attempt.**